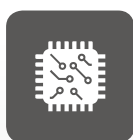
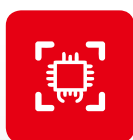


Product Selection Guide



RF IC
/Module



LoRa
Module



matter



BLE



Isolator



Sensor

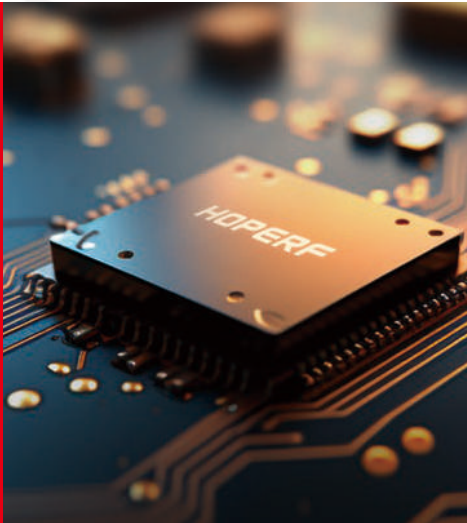
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RF IC Senior Original Manufacturer

IoT Wireless Communication System Service Vendors



HOPERF's product applications cover security, transportation, agriculture, environmental protection, fire fighting, personal consumption, public utilities and other fields.
We are committed to providing low power consumption, cost-effective, easy to expand, easy to apply the overall solution for various industries.



CONTENTS

RF



RF IC /Module

- ✓ Sub-GHz Tx/Rx/TRx IC
- ✓ Sub-GHz SoC IC
- ✓ Sub-GHz Encoder IC
- ✓ Sub-GHz Module



LoRa/LoRaWAN Module

- ✓ LoRa/LoRaWAN Module
- ✓ Other Wireless Module



matter

- ✓ Matter (over Thread)
- ✓ Matter (over WiFi)
- ✓ Matter Development



Bluetooth

- ✓ BLE IC
- ✓ BLE Module
- ✓ BLE Services

Signal Chain



Isolator IC

- ✓ Digital Isolator
- ✓ Interface Isolator
- ✓ Isolated ADC OpAmp
- ✓ Driver Isolator



Sensor

- ✓ Pressure Sensor
- ✓ Temperature & Humidity Sensor
- ✓ Customized Sensor

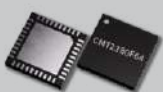
Create Values
For Clients



HOPERF

RF IC

NextGenRF® Patented Algorithm

T/R	Model No.	Modulation	Maximum Data Rate	Frequency Range	Sensitivity	Output Power	Size	Package	Remark
TRX	 CMT2310AW	OOK/2 (G)FSK /4 (G)FSK	0.2-1000kbps (4FSK) 0.1-500kbps (2FSK) 0.1-300kbps (OOK)	113-960MHz	-122dBm	20dBm	4×4×0.75mm	QFN24	4-SPI
TRX	 CMT2300AW	OOK/(G)FSK /(G)MSK	0.5-300Kbps(FSK) 0.5-40Kbps(OOK)	127-1020MHz	-120 dBm	20dBm	3*3*0.75mm	QFN(16)3x3	3+1 wire SPI Small Package Low Consumption
TRX SoC	 CMT2392F512W	OOK/2 (G)FSK /4 (G)FSK	0.2-1000kbps (4FSK) 0.1-500kbps (2FSK) 0.1-300kbps (OOK)	113-960MHz	-126dBm	20dBm	7*7*0.75mm	QFN68	CMT2310A Radio Core With Cortex M4 512KB Flash 144KB SRAM
TRX SoC	 CMT2391F128W	OOK/2 (G)FSK /4 (G)FSK	0.2-1000kbps (4FSK) 0.1-500kbps (2FSK) 0.1-300kbps (OOK)	113-960MHz	-126dBm	20dBm	7*7*0.75mm	QFN68	CMT2310A Radio Core With Cortex M4 128KB Flash 24KB SRAM
TRX SoC	 CMT2390F64W	OOK/2 (G)FSK /4 (G)FSK	0.2-1000kbps (4FSK) 0.1-500kbps (2FSK) 0.1-300kbps (OOK)	113-960MHz	-126dBm	20dBm	6*6*0.75mm	QFN48	CMT2310A Radio Core With Cortex M0 64KB Flash 8KB SRAM
TRX SoC	 CMT2380F64W	OOK/(G)FSK /(G)MSK	0.5-300Kbps(FSK) 0.5-40Kbps(OOK)	127-1020MHz	-120dBm	20dBm	5*5*0.75mm 6*6*0.75mm	QFN40 QFN48	CMT2300A Radio Core With Cortex M0 64KB Flash 8KB SRAM
TRX SoC	 CMT2380F29W	(G)FSK /(G)MSK /OOK	300Kbps	127-1020MHz	-121dBm	20dBm	5*5*0.75mm	QFN32	CMT2300A Radio Core With Cortex M0 29KB Flash 3KB SRAM
TX SoC	 CMT2168AW	OOK/(G)FSK /(G)MSK	0.5-300Kbps(FSK) 0.5-40Kbps(OOK)	27-960MHz	—	13dBm	5*5*0.75mm	QFN32	3D LF Wakeup With 8051 Core MCU Powerful AFE
TX SoC	 CMT2189DW	OOK/(G)FSK /(G)MSK	0.5-300Kbps(FSK) 0.5-40Kbps(OOK)	27-960MHz	—	13dBm	4*4*0.75mm	QFN20	With 8051 Core MCU
TX SoC	 CMT2186AW	OOK/ASK	0.5-40Kbps(OOK)	210-960MHz	—	13dBm	8.65*6*1.75 mm /9.9*6*1.75mm	SOP14 /SOP16	with 8051 Core MCU
TX SoC	 CMT2187AW	OOK/FSK /(G)FSK	0.5 - 40kbps (OOK) 0.5 - 200kbps (GFSK/FSK)	210-960MHz	—	13dBm	8.65*6*1.75 mm	SOP14	with 8051 Core MCU
TX	 CMT2156BW	OOK	0.5-40 kbps	240-960MHz	—	13dBm	3*3*0.75mm	QFN(16)3x3	With Encoder & Micro Energy Harvesting



RF IC

NextGenRF® Patented Algorithm

T/R	Model No.	Modulation	Maximum Data Rate	Frequency Range	Sensitivity	Output Power	Size	Package	Remark
TX	 CMT2157BW	OOK	0.5-40 Kbps	240-960MHz	—	13dBm	8.65*6*1.75mm	SOP14	Encoder + 14 Keys
TX	 CMT2150LW	OOK	0.5-40 Kbps	240-960MHz	—	13dBm	4.92*6*1.65mm	SOP8	Encoder + 6 Keys
TX	 CMT2119AW	(G)FSK/OOK	0.5-100Kbps (FSK) 0.5-30Kbps(OOK)	240-960MHz	—	13dBm	2.92*2.8*1.1mm	SOT23-6	1-wire Interface + TWI
TX	 CMT2119BW	(G)FSK/OOK /(G)MSK	0.5-40Kbps(OOK)	127-1020MHz	—	20dBm	3*3*0.75mm	QFN16	3+1 wire SPI
RX	 CMT2220LYW	OOK	1-5 Kbps	300-480MHz	-109dBm	—	4.92*6*1.65mm	SOP8	
RX	 CMT2220LSW	OOK	1-5 Kbps	300-480MHz	-107dBm	—	4.92*6*1.65mm	SOP8	
RX	 CMT2218BW	(G)FSK	0.5-300Kbps	300-920MHz	-120dBm	—	3*3*0.75mm	QFN16	Direct Mode
RX	 CMT2210LBW	OOK	0.1-40Kbps	300-480MHz	-113dBm	—	4.92*6*1.65mm	SOP8	Direct Mode
RX	 CMT2210LHW	OOK	0.1-40Kbps	127-1020MHz	-109dBm	—	4.92*6*1.65mm	SOP8	Direct Mode +1-wire Interface
RX	 CMT2219AW	OOK/(G)FSK /(G)MSK	0.1-100Kbps(FSK) 0.5-30Kbps(OOK)	300-960 MHz	-114dBm	—	3*3*0.75mm	QFN16	3+1 wire SPI
RX	 CMT2219BW	(G)FSK/OOK /(G)MSK	0.5-300Kbps(FSK) 0.5-40Kbps(OOK)	127-1020MHz	-120dBm	—	3*3*0.75mm	QFN16	3 wire SPI

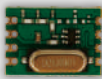
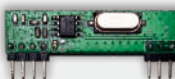


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RF Module

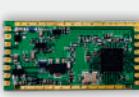
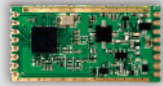
RF COB Module

Model No.		RFM300W (TRX)	RFM300CW (TRX)	RFM300HW (TRX)	RFM300PW (TRX)	RFM310W (TRX)	RFM310HW (TRX)
PHOTO							
Frequency(MHz)		315/433/868/915	433/868/915	315/433/868/915	433/868	315/433/868/915	315/433/868/915
VDD(V)		1.8~3.6	1.8~3.6	1.8~3.6	1.8~5V(1W) 7.5V(4W)	1.8~3.6	1.8~3.6
Modulation		(G)FSK/OOK	(G)FSK/OOK	(G)FSK/OOK	(G)FSK/OOK	(G)FSK/OOK	(G)FSK/OOK
Max.Data Rate (Kbps)		300/40	300/40	300/40	300/40	500/300	500/300
Current (mA)	Tx	28	28	75	800/(1W) 1400/(4W)	29	82
	Rx	7	7	7	13	9.4	9.4
Sensitivity(dBm)		-120	-120	-120	-118	-122	-122
Output Power (dBm)		13	13	20	30/36	13	20
Size(mm)		16x16x1.9	16x16x3	16x16x1.9	30x50x4.9	18x18x2.1	18x18x2.1
Interface		SPI	SPI	SPI	SPI	SPI	SPI

Model No.	RFM110W (TX)	RFM217LBW (RX)	RFM210LBW (RX)	RFM210LHW (RX)	RFM210LCFW (RX)	RFM220LSW (RX)
PHOTO						
Frequency(MHz)	315/433	315/433/868/915	315/433	315/433	315/433	315/433
VDD(V)	1.8~3.6	1.8~3.6	1.8~3.6	1.8~3.6	1.8~3.6	3.0~5.5
Modulation	OOK	OOK	OOK	OOK	OOK	OOK
Max.Data Rate (Kbps)	30	40	40	40	40	5
Current (mA)	Tx	12.4	-	-	-	-
	Rx	-	3.8	3.8	3.8	5.5
Sensitivity(dBm)	-	-113	-113	-113	-113	-109
Output Power (dBm)	13	-	-	-	-	-
Size(mm)	17.8x12.8x5	32x11x5	32x11x5	32x11x5	32x11x5	43.2x10x5
Interface	TWI	TWI	TWI	TWI	TWI	DATA

Model No.		RFM119W (TX)	RFM119BW (TX)	RFM119BPW (TX)	RFM218BW (RX)	RFM219BW (RX)	RFM219SW (RX)	RFM63W (TRX)	RFM26CW (TRX)
PHOTO									
Frequency(MHz)		315/433/868/915	315/433/868/915	433/868	315/433/868/915	315/433/868/915	315/433/868/915	868/915	434/868/915
VDD(V)		1.8~3.6	1.8~3.6	1.8~5V(1W) 7.5V(4W)	1.8~3.6	1.8~3.6	1.8~3.6	2.1~3.6	1.8~3.8
Modulation		(G)FSK/OOK	(G)FSK/OOK	(G)FSK/OOK	(G)FSK	(G)FSK/OOK	(G)FSK/OOK	FSK/OOK	(G)FSK & 4(G)FSK & ASK
Max.Data Rate (Kbps)		100/30	300/40	300/40	300	300/40	100/30	200	1000
Current (mA)	Tx	33	75	620/(1W) 1110/(4W)	—	—	—	25	90
	Rx	—	—	—	8.5	7	5.7	3	14.5
Sensitivity(dBm)		—	—	—	-120	-120	-109	-110	-109
Output Power (dBm)		13	20	30/36	—	—	—	12	20
Size(mm)		17.8x12.8x5	16x16x1.9	30x50x4.9	16x16x1.8	16x16x1.9	16x16x5	19.7x16x1.9	16x16x1.9
Interface		TWI	SPI	SPI	TWI	SPI	SPI	SPI	SPI



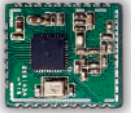
RF COB Module					RF DATA Module			
Model No.		RFM69CW (TRX)	RFM69HCW (TRX)	RFM23BPW (TRX)	RFM26CW (TRX)	HM-TRLR-SW (TRX)	HM-TRPW (TRX)	
PHOTO								
Frequency(MHz)		315/433/868/915	315/433/868/915	433/868/915	433/868/915	434/4703/868/915	315/433/868/915	
VDD(V)		1.8~3.6	1.8~3.6	5~6	1.8~3.6	2.4~3.6	2.4~3.6	
Modulation		(G)FSK/OOK	(G)FSK/OOK	FSK/OOK	(G)FSK/OOK	LoRa/(G)FSK/OOK	(G)FSK/OOK	
Max.Data Rate (Kbps)		300	300	256	1024	37.5(LoRa) 300(GFSK)	115.2	
Current (mA)	Tx	45	130	550	85	120	100	
	Rx	16	16	25	10	16	25	
Sensitivity(dBm)		-120	-120	-120	-126	-139	-117	
Output Power (dBm)		13	20	30	20	20	20	
Size(mm)		16x16x1.9	16x16x1.9	33x18x1.9	16x16x1.9	16x20x2	16x20x2	
Interface		SPI	SPI	SPI	SPI	TTL	TTL/RS485/RS232	
LoRa Module								
Model No.		RFM92LRW (TRX)	RFM90CW (TRX)	RFM95W (TRX)	RFM95PW (TRX)	RFM95CW (TRX)	RFM96W (TRX)	RFM97CW (TRX)
PHOTO								
Frequency(MHz)		434/868/915MHz /2.4GHz	150~960	869/915	869/915	137~1020	433/470	868/915
VDD(V)		1.8~3.7	1.8~3.7	1.8~3.7	5.0~6.4	1.8~3.7	1.8~3.7	1.8~3.7
Modulation		LoRa/(G)FSK/FSK	LoRa/(G)FSK	LoRa/(G)FSK/OOK	LoRa/(G)FSK/OOK	LoRa/(G)FSK	LoRa/(G)FSK/OOK	LoRa/(G)FSK
Max.Data Rate (Kbps)		62.5 kb/s(LoRa Sub-1G) 101.5 kb/s(LoRa 2.4G) 300 kb/s(GFSK)	62.5	37.5(LoRa) 300(GFSK)	37.5(LoRa) 300(GFSK)	300	37.5(LoRa) 300(GFSK)	37.5
Current (mA)	Tx	30mA @14dBm Sub-1G 110mA @22dBm Sub-1G 28mA @13dBm 2.4G	118	120	450	120	120	120
	Rx	8mA @Sub-1GHz 6.8mA @2.4GHz	10.3	10.3	10.3	10.3	10.3	12.5
Sensitivity(dBm)		-138	-137dBm@SF=12 BW=125KHz	-139	-139	-136	-139	-138
Output Power (dBm)		22 dBm@Sub-GHz 13 dBm@2.4GHz	22	20	27	20	20	20
Size(mm)		23x17.5x2	16x16x2.8	16x16x1.8	18x35.4x3.85	16x16x2.7	16x16x1.8	16x16x3
Interface		SPI	4 wire SPI	SPI	SPI	4 wire SPI	SPI	4 wire SPI
LoRa Module					2.4GHz Module			
Model No.		RFM98PW (TRX)	RFM98W (TRX)	RFM99W (TRX)	RFM68LCW (TRX)	RFM75W-S3 (TRX)	RFM75CW (TRX)	RFM75PW (TRX)
PHOTO								
Frequency(MHz)		169/433/470	433/470	2400~2483	433/470/865/915	2400~2483	2400~2483	2400~2483
VDD(V)		5.0~6.4	1.8~3.7	1.8~3.7	1.8~3.7	1.8~3.6	1.8~3.6	3.3~4.2
Modulation		LoRa/(G)FSK/OOK	LoRa/(G)FSK/OOK	LoRa/FLRC/FSK	LoRa/FSK	(G)FSK	(G)FSK	(G)FSK
Max.Data Rate (Kbps)		37.5(LoRa) 300(GFSK)	37.5(LoRa) 300(GFSK)	202(LoRa) 300(FLRC)/2000(FSK)	62.5(LoRa) 300(GFSK)	250K/1M/2M	250K/1M/2M	250K/1M/2M
Current (mA)	Tx	300/500	120	24	110	18	18	180
	Rx	10.3	10.3	8.6	8.8	16	16	21
Sensitivity(dBm)		-139	-139	-130	-129	-96	-96	-107
Output Power (dBm)		22/30	20	13	22	4	4	28
Size(mm)		18x35.4x3.85	16x16x1.8	16x16x1.8	11.5x12x2.3	16x12.8x2	16x12.8x2	33x18x2
Interface		SPI	SPI	SPI	SPI	SPI	SPI	SPI



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RF Module

SoC COB Module

		RFM14W (TRX)	RFM23A020W (TRX)	RFM25A12W (TRX)	RFM380F64W (TRX)	RFM390F64W (TRX)
PHOTO						
Frequency(MHz)		433/190/868/915	868/915	433/868/915	433/868/915	433/868/915
VDD(V)		1.8~3.6	1.8~3.8	1.8~3.6	1.8~3.6	1.8~3.6
Modulation		(G)FSK	(G)FSK/OQPSK/OOK	(G)FSK/OQPSK/OFDM	(G)FSK/OOK	(G)FSK/OOK
Max.Data Rate (Kbps)		500	2M	2048	300	1000
Current (mA)	Tx	100	77	76.6	80	80
	Rx	11	5	6.3	12	11
Sensitivity(dBm)		-115	-118	-125	-120	-122
Output Power (dBm)		20	20	16	20	20
Size(mm)		20x23x2.0	16.5x19x2.0	19x24x1.0	16x18x2.0	19x19x1.0
Interface		SoC	SoC	SoC	SoC	SoC

WiFi Module

LoRaWAN Module

		HM-WF7201W	RFM6601	RFM6601SE
PHOTO				
Frequency(MHz)		2.4GHz	470/868/915 MHz	470/868/915 MHz
VDD(V)		2.7~3.6V	1.7~3.7V	1.7~3.7V
Modulation		WiFi + BLE	LoRa	LoRa
UART Baund Rate		115.2Kbps	115.2Kbps	115.2Kbps
Air Rate		TBD	62.5 kb/s(LoRa) 300 kb/s(GFSK)	62.5 kb/s(LoRa) 300 kb/s(GFSK)
Current (mA)	Tx	TBD	108mA(470MHz) 120mA(868/915MHz)	122mA(868/915MHz)
	Rx	TBD	<10mA	<9mA
Sleeping Current	Rx	TBD	1.3uA(Typ.)	1.5uA(Typ.)
Sensitivity (dBm)		-99dBm	-138dBm	-138dBm
Output Power (dBm)		+20dBm	+22dBm	+22dBm
Size(mm)		20.3x15.8x2.6mm	16x18x2.8mm	19x19x2mm
Interface		UART/HSPI/I2C/I2S/GPIO/PWM	UART	UART

**HOPERF** **matter**

Matter Over Thread Module					Matter Over WiFi Module	
	HM-MT2401	HM-MT2401B	HM-MT2401C	HM-MT2401 DK Board	HM-MT7201	HM-MT7203
Photo						
Frequency	2.4GHz	2.4GHz	2.4GHz	2.4GHz	2.4GHz	2.4GHz
Voltage	1.71~3.8V	1.71~3.8V	1.71~3.8V	1.71~3.8V	3.3V	3.3V
Size	22x15.8x2.6mm	22x15.8x2.6mm	12.7x10x2.6mm	79x48x15mm	20.3x15.8x2.6mm	22.3x15.8x2.6mm
TX Power	19.5 dBm	10 dBm	10 dBm	19.5 dBm	20 dBm	20 dBm
RAM	256KB	256KB	256KB	256KB	288KB	640KB
FLASH	1.5MB	1.5MB	1.5MB	1.5MB	2MB	4MB
Ultra-Low Power	○	✓	✓	○	○	✓
Mini Size	○	○	✓	○	○	○
High TX power	✓	○	○	✓	✓	✓
Supported Protocols						
Rapid Development	✓	✓	✓	✓	✓	✓
Built-in PCB Antenna	✓	✓	○	✓	✓	✓
Customized Size	✓	✓	✓	○	✓	✓
Peripherals GPIO	✓	✓	✓	✓	✓	✓
OTA in Internal Flash	✓	✓	✓	✓	○	✓

6

HOPERF **matter**

Matter is a unified communication language for smart homes, residences, and commercial buildings, simplifying connectivity and interaction among various smart devices. This new standard lays a solid foundation for the future of smart home technology, representing the next generation in the field.

csa connectivity standards alliance
CERTIFICATE OF MEMBERSHIP

Shenzhen Hope Microelectronics Co., Ltd.

member since year 2023

is a member in good standing with the Connectivity Standards Alliance. The Alliance is an association of worldwide companies working together to create and develop open global standards that enable innovative wireless communication for the Internet of Things in connected devices for consumer, commercial, and industrial industries.

*Tobin Richardson*Tobin Richardson
President & CEO, Connectivity Standards Alliance

NOT VALID FOR OTHER USES WITHOUT PERMISSION

CSA Participant Member**Cross-platform compatibility**

Matter prioritizes cross-platform compatibility, enabling seamless connectivity and interaction among diverse device brands and types. It focuses on breaking down platform barriers for increased collaboration.

**Open and unified standards**

Matter's open standard solves compatibility issues among diverse smart devices, allowing easy integration of devices from different brands into a unified smart home ecosystem.

**Security and privacy protection**

Matter prioritizes the security and privacy of smart home devices, implementing advanced security protocols such as encrypted communication and authentication measures to ensure user data safety and privacy protection.

**Broad development potential**

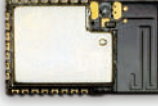
Matter, backed by a powerful alliance that includes tech giants such as Apple, Amazon, Google, Samsung, enjoys widespread industry recognition and support, fostering its future development and promotion.

BLE Module

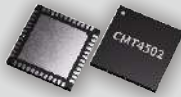
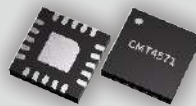
BLE Module

	HM-BT4531	HM-BT4531E	HM-BT4531B	HM-BT4502
Photo				
Bluetooth Core	5.2	5.2	5.2	5.0
Flash Memory	FLASH256KB RAM48KB	FLASH256KB RAM48KB	FLASH256KB RAM48KB	FLASH512KB RAM138KB
GPIO	5	17	17	5
PHY	1M/2M/125K/500K	1M/2M/125K/500K	1M/2M/125K/500K	1M/2M
BLE MESH Networking	YES	YES	YES	N/A
TX Power Max.	+6dBm Max.	+20dBm Max.	+6dBm Max.	+8dBm Max.
Sensitivity(dBm)	-96dBm@1Mbps -103dBm@125Kbps	-98dBm@1Mbps -105dBm@125Kbps	-96dBm@1Mbps -103dBm@125Kbps	-97dBm@1Mbps
Current (mA)	Tx	4.2mA@0dBm	130mA@20dBm	4.2mA@0dBm
	Rx	3.8mA@3.3V	2.6mA@3.3V	3.8mA@3.3V
Sleeping Current	1.4uA	< 5uA	1.4uA	13uA
VDD(V)	1.8~3.6V	1.8~3.6V	1.8~3.6V	1.8~3.6V
Size(mm)	17x12.5x2.6mm	13x21x2.8 mm	19.02x11.22x2.6mm	17x12.5x2.5mm
Certification	BQB, FCC/CE/IC/SRRC	N/A	N/A	BQB, FCC/CE/IC/SRRC
Default Interface	UART	UART	UART	UART
UART Baud Rate	115200bps	115200bps	115200bps	115200bps

BLE Module

	HM-BT2701	HM-BT2401DA	HM-BT220X	HM-BT210X
Photo				
Bluetooth Core	5.4	6.0	5.2	5.2
Flash Memory	FLASH768KB RAM64KB	FLASH1024KB RAM128KB	FLASH352KB 512KB RAM32KB	FLASH768KB RAM64KB
GPIO	14	26	14	16
PHY	1M/2M/125K/500K	1M/2M/125K/500K	1M/2M/125K/500K	1M/2M/125K/500K
BLE MESH Networking	YES	YES	LPN Only	YES
TX Power Max.	+8dBm Max.	+10dBm Max.	0dBm /+6dBm Max.	+10dBm/+20dBm Max.
Sensitivity(dBm)	-99.2dBm@1Mbps -106.9dBm@125Kbps	-97.6dBm@1Mbps -105.7dBm@125Kbps	-98.9dBm@1Mbps -106.7dBm@125Kbps	-97.5dBm@1Mbps -104.9dBm@125Kbps
Current (mA)	Tx	4.1mA@0dBm	5.0mA@0dBm	4.1mA@0dBm
	Rx	3.6mA@3V	4.4mA@3.3V	3.6mA@3V
Sleeping Current	1.6uA	1.3uA	1.4uA	5.0uA
VDD(V)	1.8~3.8V	1.71~3.8V	1.71~3.8V	1.71~3.8V
Size(mm)	12x17x2.1mm	29x26x2.6mm	17x12x2.1mm	20x12x2.6mm
Certification	N/A	N/A	BQB, FCC/CE/IC/SRRC	CE
Default Interface	UART	UART	UART	UART
UART Baud Rate	115200bps	115200bps	115200bps	115200bps

**HOPERF****BLE IC****BLE IC**

	CMT4531	CMT4502	CMT4571
Photo			
Bluetooth Core	5.2	5.0	5.0
Processor	Cortex-M0 64MHz	Cortex-M0 48MHz	Cortex-M0 64MHz
Flash Memory	FLASH256KB RAM48KB	FLASH512KB RAM138KB	FLASH512KB RAM56KB
GPIO	21	19(QFN32) 33(QFN48)	11 (QFN20)
Sleeping Current	1.4uA	13uA	3.5uA
Sensitivity(dBm)	-96dBm	-97dBm	-99 dBm
TX Power Max.	+6dBm Max.	+10dBm Max.	+8dBm Max.
PHY	1M/2M/125K/500K	1M/2M	1M/2M/125K/500K
Size(mm)	QFN32(4x4mm)	QFN32(4x4mm) QFN48(7x7mm)	QFN20(3X3mm)

8

**HOPERF provides a wide range of BLE system solution services****BLE IC & Module****BLE IC Customized Development****BLE IC System
Cooperation Development****BLE IC Patent Licensing****Bluetooth Module
Software Customization****BLE IC/Module Burn-In Service****One-stop BLE
Solutions****BLE Supports & Training****Application**

Medical



Sports



Wearable



Entertainments



Smart Home



CAR Electronics



Pets Training



Smart Toys



Isolator

Part Number	Channels	Forward Channels	Reversed Channels	Default Output State	Maximum Data Rate	Operating Voltage	Isolation Voltage	Surge Voltage	CMTI	Package
Digital Isolator										
CMT8020N0L	2	2	0	L	10Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8020N1L	2	2	0	H	10Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8021N0L	2	1	1	L	10Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8021N1L	2	1	1	H	10Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8022N0L	2	1	1	L	10Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8022N1L	2	1	1	H	10Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8120N0	2	2	0	L	150Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8120N1	2	2	0	H	150Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8121N0	2	1	1	L	150Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8121N1	2	1	1	H	150Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8122N0	2	1	1	L	150Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8122N1	2	1	1	H	150Mbps	2.5-5.5V	3.75kV	5.3kV	200kV/ μ s	SOP8
CMT8120W0	2	2	0	L	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW8
CMT8120W1	2	2	0	H	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW8
CMT8121W0	2	1	1	L	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW8
CMT8121W1	2	1	1	H	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW8
CMT8122W0	2	1	1	L	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW8
CMT8122W1	2	1	1	H	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW8
CMT8040N0	4	4	0	L	150Mbps	2.5-5.5V	3.75kV	5.3kV	150kV/ μ s	SOP16
CMT8040N1	4	4	0	H	150Mbps	2.5-5.5V	3.75kV	5.3kV	150kV/ μ s	SOP16
CMT8041N0	4	3	1	L	150Mbps	2.5-5.5V	3.75kV	5.3kV	150kV/ μ s	SOP16
CMT8041N1	4	3	1	H	150Mbps	2.5-5.5V	3.75kV	5.3kV	150kV/ μ s	SOP16
CMT8042N0	4	2	2	L	150Mbps	2.5-5.5V	3.75kV	5.3kV	150kV/ μ s	SOP16
CMT8042N1	4	2	2	H	150Mbps	2.5-5.5V	3.75kV	5.3kV	150kV/ μ s	SOP16
CMT8040W0	4	4	0	L	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW16
CMT8040W1	4	4	0	H	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW16
CMT8041W0	4	3	1	L	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW16
CMT8041W1	4	3	1	H	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW16
CMT8042W0	4	2	2	L	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW16
CMT8042W1	4	2	2	H	150Mbps	2.5-5.5V	5kV	7kV	200kV/ μ s	SOW16



Part Number	Channels	Forward Channels	Reversed Channels	Default Output State	Maximum Data Rate	Operating Voltage	Isolation Voltage	Surge Voltage	CMTI	Package
CMT8260W0	6	6	0	L	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8260W1	6	6	0	H	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8261W0	6	5	1	L	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8261W1	6	5	1	H	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8262W0	6	4	2	L	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8262W1	6	4	2	H	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8263W0	6	3	3	L	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8263W1	6	3	3	H	150Mbps	2.5-5.5V	5kV	10kV	250kV/ μ s	SOW16
CMT8260N0	6	6	0	L	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8260N1	6	6	0	H	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8261N0	6	5	1	L	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8261N1	6	5	1	H	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8262N0	6	4	2	L	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8262N1	6	4	2	H	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8263N0	6	3	3	L	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16
CMT8263N1	6	3	3	H	150Mbps	2.5-5.5V	3.75kV	10kV	250kV/ μ s	SOP16

Part Number	Transmission Speed	Isolation Voltage	CMTI	Surge Class	ESD Rating	Operating Voltage	Clock	Temperature Range	Package
Isolation Interface I2C									
CMT8100N	2MHz	3.75kV	150kV/ μ s	7kVpk	8KV	2.5-5.5V	Bidirectional	-40°C~125°C	SOP8
CMT8101N	2MHz	3.75kV	150kV/ μ s	7kVpk	8KV	2.5-5.5V	Unidirectional	-40°C~125°C	SOP8
CMT8100WH	2MHz	5kV	150kV/ μ s	7kVpk	8KV	2.5-5.5V	Bidirectional	-40°C~125°C	SOW8
CMT8101WH	2MHz	5kV	150kV/ μ s	7kVpk	8KV	2.5-5.5V	Unidirectional	-40°C~125°C	SOW8

Isolator Interface CAN									
CMT1050	1Mbps	5kV	150kV/ μ s	8kVpk	8KV	2.5-5.5V	/	-40°C~125°C	SOW8
CMT1042	5Mbps	5kV	150kV/ μ s	8kVpk	8KV	2.5-5.5V	/	-40°C~125°C	SOW8
CMT1050W	1Mbps	5kV	150kV/ μ s	8kVpk	8KV	2.5-5.5V	/	-40°C~125°C	SOW16
CMT1052W	5Mbps	5kV	150kV/ μ s	8kVpk	8KV	2.5-5.5V	/	-40°C~125°C	SOW16
CMT1042W	5Mbps	5kV	150kV/ μ s	8kVpk	8KV	2.5-5.5V	/	-40°C~125°C	SOW16

Part Number	Transmission Speed	Isolation Voltage	CMTI	Surge Class	ESD Rating	Operating Voltage	Clock	Temperature Range	Package
Isolator Interface RS-485									
CMT83085	12Mbps	5kV	150kV/ μ S	10kVpk	8KV	2.5-5.5V	half-duplex	-40°C~125°C	SOW16
CMT83086	12Mbps	5kV	150kV/ μ S	10kVpk	8KV	2.5-5.5V	full-duplex	-40°C~125°C	SOW16
CMT83089C	500Kbps	5kV	150kV/ μ S	10kVpk	8KV	2.5-5.5V	half-duplex	-40°C~125°C	SOW16

Part Number	Isolation Voltage	Linear Input Voltage	Input Mode	Output Mode	CMTI	Temperature Range	Package
Isolated ADC/OP-AMP							
CMT1306M05	5kV	-50~50mV	Differential	Numerical	150kV/ μ S	-40°C~125°C	SOW8
CMT1306M25	5kV	-250~250mV	Differential	Numerical	150kV/ μ S	-40°C~125°C	SOW8
CMT1300D05	5kV	-50~50mV	Differential	Differential	150kV/ μ S	-40°C~125°C	SOW8
CMT1300D25	5kV	-250~250mV	Differential	Differential	150kV/ μ S	-40°C~125°C	SOW8

Part Number	Isolation Voltage	Drive Current	Low-Voltage Protection	CMTI	Temperature Range	Driver	Package
Isolated Drive							
CMT8602A-N	3.75kV	4A-6A	6V	150kV/ μ S	-40°C~125°C	Half-Bridge	SOP16
CMT8602A-K	5.75kV	4A-6A	6V	150kV/ μ S	-40°C~125°C	Half-Bridge	SOW14
CMT8602B-N	3.75kV	4A-6A	9V	150kV/ μ S	-40°C~125°C	Half-Bridge	SOP16
CMT8602B-K	5.75kV	4A-6A	9V	150kV/ μ S	-40°C~125°C	Half-Bridge	SOW14
CMT8602C-N	3.75kV	4A-6A	13V	150kV/ μ S	-40°C~125°C	Half-Bridge	SOP16
CMT8602C-K	5.75kV	4A-6A	13V	150kV/ μ S	-40°C~125°C	Half-Bridge	SOW14
CMT8603B-WF	5.75kV	5V	9V	150kV/ μ S	-40°C~125°C	Optocoupler Compatible	SOW6
CMT8603C-WF	5.75kV	5A	13V	150kV/ μ S	-40°C~125°C	Optocoupler Compatible	SOW6



Excellent
Performance



High Reliability



Anti-Interference



Flexible Delivery

Application



Automation



Equipment



Multi-channels



Electrical



Power Transmission



Base Station



Telecommunications



Smart Metering



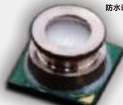
Security



Photovoltaic

**HOPERF**

Pressure Sensor

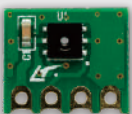
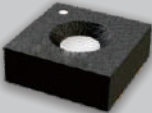
Model NO. & Photo	Pressure Range	Relative Accuracy	Temp Range & Accuracy	Working Voltage	Sleep Current	Size	Package	Interface
 HPS700A-ZM	0~ 1600Kpa	±2.0 Kpa	-40 ~ 85℃	1.8~3.6V	<0.1uA	3.8*3.6*1.15 mm	LGA8	I ² C
 HPS27W	300~ 1200Hpa	±1.0 Hpa	-40 ~ 85℃	1.7~3.6V	<0.5uA	2.7*2.7*1.7 mm	LGA10	I ² C
 HP5834	300~ 1200Hpa	±2.0 Hpa	-40 ~ 85℃	1.8~3.6V	<0.1uA	4.5*4.5*3.0 mm	DFN6	I ² C
 HP206N	300~ 1200Hpa	±2.0 Hpa	-40 ~ 85℃	1.8~3.6V	<0.1uA	6.8*6.2*3.1 mm	DFN6	I ² C
 HP209-XXXG-T	0~10Kpa 0~200Kpa	±1% @FS	-40 ~ 85℃	1.8~3.6V	<0.1uA	9.0*9.0*11.9 mm	DFN6	I ² C
 HP209-XXXG-W	0~10Kpa 0~200Kpa	±1% @FS	-40 ~ 85℃	1.8~3.6V	<0.1uA	9.0*9.0*4.4 mm	DFN6	I ² C
 6862I	300~ 1200Hpa	1.0 Hpa	-40 ~ 85℃	1.7~3.6V	<0.5uA	6.8*6.2*3.1 mm	LGA8	I ² C/SPI
 HPSXXXGSR	±10Kpa ±40Kpa ±100Kpa ±200Kpa	±1% @FS	-40 ~ 85℃	3.0~5.5V	<0.2uA	7*10.6*10 mm	SOP6	I ² C
 PTM00X	0~100Bar	±0.5% @FS	-40℃~125℃	2.7V~5.5V	-	G1/4 1/4NPT R1/4 *Customizable	-	4~20mA 0.5~4.5V I ² C *Customizable



HOPERF

Model NO. & Photo	Pressure Range	Relative Accuracy	Temp Range & Accuracy	Working Voltage	Sleep Current	Size	Package	Interface
 HP303S	300~1200Hpa	±1.0 Hpa	-40 ~ 85°C	1.7~3.6V	<0.5uA	2.5*2*0.95 mm	LGA8	I ² C
 HP303B	300~1200Hpa	±1.0 Hpa	-40 ~ 85°C	1.7~3.6V	<0.5uA	3.8*3.6*1.2 mm	LGA8	I ² C
 HP203N	300~1200Hpa	±2.0 Hpa	-40 ~ 85°C	1.8~3.6V	<0.1uA	3.8*3.6*1.2 mm	LGA8	I ² C

Temperature & Humidity Sensor

Model NO. & Photo	Temp Range	Temp Resolution	Humidity Range	Humidity Resolution	Working Voltage	Sleep Current	Size	Interface
 TH10	-40~125°C	±0.1°C	0~100%	±1.5%	2.4~5.5V	0.2uA	10.8*9.1*1.8 mm	I ² C
 TH09C	-40~100°C	±0.15°C	0~100%	+2.0%	1.71~3.6V	0.04uA	2.0*2.0*0.75 mm	I ² C
 T09	-40~125°C	±0.2°C	N/A	N/A	1.7~3.6V	0.1uA	WLCSP	I ² C

Custom Types

Measurement Range

Application

Picture

Gauge Pressure Sensor

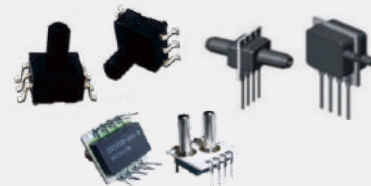
15kpa ~ 3.5MPa

Tire Pressure Monitoring, Security, Air Conditioning, Industrial Control, Industrial Instrument, Etc.

Differential Pressure Sensor

2kpa ~ 3.5MPa

Flow, Level, Medical, Leak Detection, Etc.



Application



Smart Meter



UAV



Sports Watch



Weather Station



Medical Machine



Smart-Home



Air Pumps



Health Machine

Development Tools

HopeDuino™



HOPERF Arduino IDE
Development KIT



CMT2380F64 EB



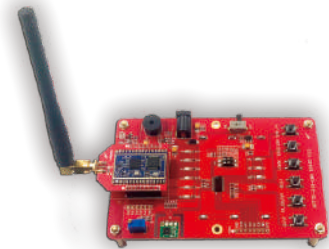
RF-EB



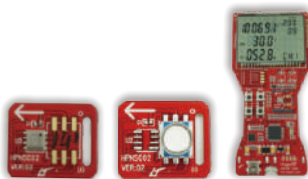
USB Programmer



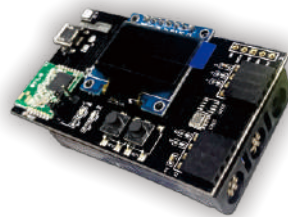
Arduino IDE
Universal KIT



LoRa Demo Board



HP20x Demo Board



Sensor EVB



Offline Writer

More Applications



Smart Home

Household Automation
Security Monitoring
Kitchen & Bath Security
Monitoring
Home Information Center
Home Environment
Controlling



Landscape Sites

To provide facilities of security,
lighting, flow control, guidance,
weather...etc for scenic spots
To provide basis for the protection of
scenic areamart Animal Farm



Intelligent Nursing Home Management

Monitor the physical condition of the elderly
Whether the elderly is out of the range
Provide a good environment for the elderly
Provide timely information to kinsfolk



Intelligent Streetlight

Lighting Automation Controlling,
Single Lamp
Switch & Brightness Control
Power Supply Circuit Controlling
& Circuit
Detection



Smart Metering

Smart Water Meter,
Smart Gas Meter,
Smart Heat Meter
Pressure Data, Pipe
Leakage Detection
Temperature Data,
Abnormity Warning
Remote Parameter
Configuration



Intelligent Hospital Infusion Management

Monitor Infusion Situation
Whether the patient is out of
the range
Remind the nurse in time
Provide Mega Data for
Hospital Infusion



Intelligent Photovoltaic Management

Environmental Parameter
Detection
Photovo Itaic Power
Generation Efficiency Detection
Inverter Condition Detection
Troubleshooting



Smart Agriculture Shed

Light Intensity
Detection, CO2 Detection, Humiture Detection
Soil Temperature-
moisture and Fertilizer Detection
Spray & Irrigation Controlling
Sunshade & Aerofluxus Controlling



Smart Manhole Cover

Intelligent Lock, Personal
Identification
Position Demanding,
Manhole Cover Stealing Warning
Water-level Monitoring,
Flood Warning
Water Quality Sampling,
Biochemical Analysis



Farming monitoring

Environmental Status Detection
Environmental Control Equipment
Electronic Fence, Leaving Alarm
Animal Physical Condition Detection



Intelligent Aquiculture Farms

Water Temp & O2
Concentration Detection
Water Quality Parameter Detection
Water Pump Detection
& Water Quality Adjustment
Fish Growth Density Detection



Cultural Relics Safety Protection

Provide high-tech security for cultural relics,
monuments and museums
Combined remote control lighting, alarm,
camera
Intelligent Control of Scenic Spot



Intelligent Pipeline System

Pipeline Hydraulic &
Pressure Detection,
Wireless Warning in time
Remote Valve Controlling
Monitoring Safety Controlling



Intelligent City

Remote Controlling
Operation Analysis,
Proactive Maintenance
Tumble Recognition,
Abnormity Warning
Ads Service



Intelligent Outdoor Environmental Detection

Base Station Signal Coverage, Environmental
Monitoring, Advertising Display, Charging Pile



Central Air-conditioning Control

Remote Controlling
Intelligent Temperature Controlling
To achieve the energy saving and
efficient
operation of the central air-
conditioning system
Elevator Net-connected



Intelligent Parking

Collecting Parking
Information
Parking Guidance
Back/Finding Cars
Parking Charge



Intelligent Hillside Soil and Water Monitoring

To monitor mountain
slope moisture for soil-water protection
All-time surveillance of important slopes,
tunnels, riverbanks, reservoirs, railways, etc



Vehicle Engineering Management

Fuel Consumption
Monitoring, Operation State Monitoring
Operational Monitoring,
Intelligent Scheduling
Track Playback, Speed Limit Area



Intelligent Agriculture

Soil Moisture Monitoring,
Agricultural Environmental
Monitoring
Meteorological Monitoring,
Air Quality Monitoring
Water Conservancy and
Hydrology, Water Quality
Detection



Intelligent Wastebin

Intelligent sending full/empty flag
of rubbishes for better controlling
Air cleaning for better environment
Sensor switch for better experience
of dumping wastes



SHENZHEN HOPE MICROELECTRONICS CO., LTD (hereinafter referred to as "HOPERF"), with a registered capital of RMB 60 million, is a national high-tech enterprise which focusing on wireless RF chips and sensors. HOPERF has now developed into a complete industrial company that integrates ASIC chip design, MEMS sensor chip design, packaging test calibration technique, and application developing and service. HOPERF has an analog and digital hybrid radio frequency chip "NextGenRF" algorithm, PlutoIoT/PlutoWAN protocol, and silicon pressure-resistant sensor chip design technology; the main features based on Sub-GHz RF chips and extended wireless application modules; various types of MEMS sensors, as well as related IoT application solutions, etc.

HOPERF products lines include wireless RF IC(based on Sub-GHz), IoT modules based on the IC, and signal-chain products including sensors and isolators, etc. also HOPERF provides IoT-Related products such as BLE module, WiFi module, Matter module etc., covering information collection, information transmission and signal processing, with broad expansion applications.



After many years of steady development, HOPERF started from the market service-oriented sales of the market, deepened customer needs for targeted development, provided technical research and development system services, and went to R&D with production. HOPERF have deep experience precipitation and good service reputation in the field of RF IC industry. Aiming at the various scenarios of customers in product development, design and mass production, HOPERF provides customers with a series of precise services including products data, selection consulting, free samples, testing tools, technical support, and quality control guarantee. The multi-fields experience flexible industry application projects help customers to made massive cases implemented quickly and successfully.

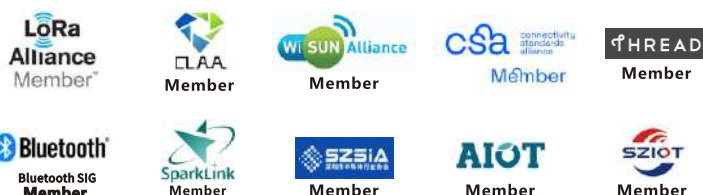
The products of HOPERF have covered dozens of countries to serve global customers, and have been applied to increasingly widely use in consumer electronics, security products, data collection, outdoor sports, electronic navigation, industrial control & automatic, environmental monitoring, medical health, smart home, and other fields. Adhere to independent research and development, provide products & services such as RF chips and sensors for business, consumption and industrial applications, and actively promote the gradual and controllable localization pace of China's integrated circuit industry.

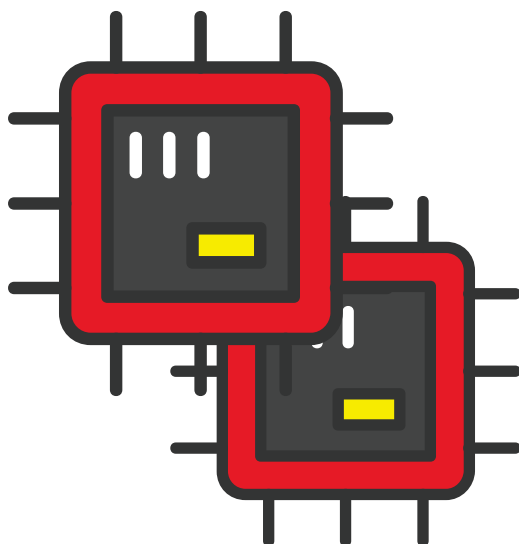
In the future, HOPERF strives to upgrade core technologies, achieve a leading position in the certain RF IC and Signal Chain industry through advanced design, superior performance, and reliable quality products that promote our customers benefits. HOPERF is committed to improving continuous progress of the industry, actively create social value, and achieve sustainable healthy developments.



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